AGREEMENT ANALYZER

Tells a borrower what a loan will really cost, before they sign it

A loan advertised at **24 per cent** really costs **42.8 per cent** once one fee of Rs. 4,500 is counted.

WHAT IT DOES

- Reads a PDF, a phone photo, a scan, a Word file or text, in 9 languages.
- Checks it is really an agreement, and works out which kind.
- Finds the risky clauses with 27 rules and paints a box on every risky line.
- Explains each one in plain English or Hindi, and reads it aloud.
- Works out the real monthly instalment and the real yearly cost with fees.
- Answers the borrower's own questions about the document.
- Writes the complaint letter and shows where to escalate.

HOW IT ANSWERS QUESTIONS, WHICH IS THE RAG PART

- The document is cut into overlapping passages; each becomes a list of 384 numbers.
- A question becomes 384 numbers too, and the five closest passages are found.
- The answer is written from those passages only, so it cannot invent a clause.
- The model runs on the server, so nothing is sent away and no key is needed.

BUILT WITH

- TypeScript on Node.js 24, Next.js 16, React 19, Tailwind
- pdfjs, tesseract.js, sharp, mammoth, canvas rendering
- all-MiniLM-L6-v2 embeddings on the ONNX runtime, run locally
- 27 rule risk engine, true yearly rate by bisection
- AES-256-GCM on every file, record and search index
- Clerk sign in, zod, helmet, rate limits, owner only reads
- Docker, GitHub Actions, CodeQL, Dependabot, secret scanning

MEASURED

- **91.7%** document type, by cross validation
- **80.0%** rules judging harm, on 40 labelled clauses
- **45.0%** a learned classifier on the same 40, so rules decide risk
- **93%** confidence reading from photographs
- **37** automated tests, run on every change
- **0** known vulnerabilities in the server
- Working public demo, not only a local build

STATUS

Report, working code and demonstration slides are all complete. Everything runs with no API key and no paid service, so it can be demonstrated anywhere.